

OligoTox-Coagulopathy — Methodology

Materials and methods · NIH/NCATS Oligonucleotide Toxicity Open Data Challenge, Phase 2

Materials and methods for the coagulation-toxicity dataset, NIH/NCATS Oligonucleotide Toxicity Open Data Challenge, Phase 2.

1. Design

This is a **curated dataset**. No wet-lab experiment was performed. The methods below are therefore of two kinds, kept strictly apart: **(a)** the experimental methods of the source studies, recorded as each source states them, and **(b)** the curation methods used to find, extract, verify and harmonise them. Conflating the two would let curation choices masquerade as experimental fact.

Snapshot: **218 oligonucleotides · 2,685 measurements · 1,039 per-position modification records · 100 sources.**

2. Scope — what counts as this endpoint

Included: clotting times (aPTT, PT/INR, TT, ACT), fibrinogen, D-dimer, thrombin generation, coagulation-factor and antithrombin activity, anti-Xa/anti-IIa activity, and bleeding or thrombotic outcomes a source attributes to a coagulation defect.

Excluded, deliberately: **platelet count alone**. Thrombocytopenia is a separate Challenge endpoint with its own dossier at `../thrombocytopenia.md`. A

platelet-related row appears here only where the source ties it to a cascade readout, and says so in `notes`. This boundary was set before extraction, after finding that the repository's largest existing store of "coagulation-adjacent" text — 16 hits on one page of a review already held — is entirely thrombocytopenia and belongs to that endpoint.

Both **unintended coagulopathy** and **on-target anticoagulant/procoagulant pharmacology** are in scope, on separate flags, because the second is where nearly all published per-compound clotting numbers live and because bleeding risk from an on-target anticoagulant is a genuine safety signal. They are never pooled: see README.

3. Source identification

Eight independent search axes were run in parallel — mechanism/in-vitro assays; clinical trials; nonclinical toxicology; siRNA and LNP; aptamers; regulatory labels; patents; reviews used as a citation map. Each axis was required to **retrieve and open** a candidate and quote the coagulation passage before reporting it; nothing entered the work-list on the strength of a title or a search summary. 96 raw candidates were merged to 73 distinct sources, then triaged into an extraction work-list ranked by whether the source gives per-compound numbers, per-compound qualitative statements, or class-level prose only.

Two results of that pass are worth recording because they are negative:

- **The project's only previously recorded lead was closed.** `coagulopathy.md` named Crooke et al. (2016), **Mol Ther** 24(10):1771–1782, as the single lead for this endpoint, on the strength of its title. It was retrieved and read: it does not report the per-compound coagulation values the title implies. It is recorded as context, not as a source of rows.

- Three sources whose granularity two axes disagreed about were resolved **downward**, to qualitative, because their values live only in figure panels.

4. Retrieval

All documents were retrieved directly; no value in this dataset comes from a search-engine summary. This is a deliberate break from the sibling kidney dataset, which was built under a network policy that blocked outbound fetch and whose own validation file concluded that its search-derived clinical rows had produced a provenance/outcome confound. Routes used:

Route	Sources
PMC / Europe PMC open-access JATS full text	27
NLM DailyMed SPL XML (US prescribing information)	8
NCBI eutils PubMed abstract (where no open full text exists)	8
USPTO grant text	7
Other staged full text and supplementary files	25

Every retrieved document is committed to `sources/documents/`, so each row's evidence can be re-read from the repository without network access. One supplementary PDF could not be retrieved; the 14 rows citing it are flagged and excluded from source verification rather than quietly passed.

5. Extraction

Sources were divided into 14 bundles by compound family and document type, and read individually. Extraction rules, enforced by the output contract:

- Every measurement carries a **verbatim quote** and an **exact locus** (table number, figure panel, section heading, label section, page). A row that could not be quoted was not written.
- **No value was read off a figure.** Where a number exists only in a plotted panel, the row carries `NOT_REPORTED` with `readout_is_qualitative = TRUE`. This cost real yield — one source's entire PT/aPTT time course is a figure — and the loss is recorded rather than recovered by pixel-reading.
- **Per-position chemistry is transcribed, never modelled.** A `modifications` row exists only where the source publishes position-resolved chemistry, and each row records in `basis` how the position was determined, normally the source's own case legend quoted verbatim.
- Severity is recorded **in the source's own words**; no grade was assigned during

extraction.

6. Oligonucleotide identity, purity and characterisation

Until 2026-10-03 this section reported `purity_pct` as `NOT_REPORTED` for every compound and blamed the literature. **That was wrong and the error was ours:** earlier extraction read the **clinical** sections of the EMA assessment reports and FDA integrated reviews already held in `sources/documents/` and never opened their Quality/CMC sections. Nothing had to be acquired. Recovery used eleven parallel extraction passes, one per compound programme, each checked by an independent pass that string-matched every quote against the cited file and every purity value against a printed purity context. Eight verified clean; three reported metadata defects (a spliced quote, two page misattributions, one unsupported claim about redaction markers), each corrected in `sources/characterisation.json` with the correction recorded in the record. Those checks now run inside `verify_against_sources.py` on every build.

Two distinctions carry the result. A public EPAR **names** every purity and impurity test and **withholds** the numeric limit — volanesorsen's active-substance specification lists purity, specified and unspecified impurities and total degradation products, all by IP-HPLC-UV-MS, with no limit printed for any. That is `purity_limits_redacted = TRUE` with `purity_pct` left `NOT_REPORTED` on 12 compounds: a withheld limit is a finding, never a value. And a purity value belongs to a **lot** — fitusiran 98.5% (lot P07916), olezarsen 91.3% (drug substance CA678354-002), tofersen 90% (TA666853-008) and 94% (TA666853-001), all tested batches. Tofersen therefore carries no single `purity_pct`; both values sit in `purity_batches` and the basis says why. A drug-substance specification is likewise never spread across batches, and QC fails the build if a compound with several lot purities reports a single one.

Coverage, against the denominator that matters — the 38 compounds actually dosed in people, with the whole roster for comparison:

	all 218	38 dosed in participants
sequence as printed	121	14
position-resolved chemistry	52	11
purity value	2	2
purity test named, limit withheld	12	12
purity method named	46	11
identity confirmation	32	16

Counterion is recorded for 17 of the 38, purification or manufacture for 14, impurity classes for 12 and other characterisation tests for 16. Against the previous release that is 0→2 purity values, 1→11 named methods, 3→16 identity confirmations and 0→14 purification records on the clinically dosed subset. It remains **thin**, and the two worst gaps are untouched: only 14 of 38 have a printed sequence and only 11 have position-resolved chemistry.

The recovered methods are consistent across sponsors. **Identity** — accurate mass by MS within an IP-HPLC-UV-MS run; sequence confirmation by thermal melting temperature; failure-sequence analysis of crude material by IP-HPLC-TOF-MS; structure elucidation by ¹H, ¹³C and ³¹P NMR with high-resolution ESI-TOF. **Purity** — one IP-HPLC-UV-MS method determines assay, purity and impurities together, with impurities specified as *groups* rather than individual components. **Other attributes** — counterion by ICP-OES, elemental impurities by ICP-MS, water by Karl Fischer, residual solvents by GC, endotoxin and microbial limits by Ph. Eur. One fact deserves emphasis: a 20-mer full-phosphorothioate is a mixture of 2¹⁹ (524,288) diastereoisomers, no individual isomer above roughly 0.0018% of the total. ****"The sequence" and "the molecule" are not the same thing for a phosphorothioate**, and a model treating a PS sequence as one species is modelling a mixture.**

Identity here means the printed sequence plus its per-position chemistry. QC checks that every declared length equals the actual string (plus any documented terminal residue) and that every modification row's nucleobase matches the sequence at that position. No purity value was estimated, inferred from a synthesis platform, or carried across compounds; regulatory-derived and publication-derived values are held in separate columns.

7. Harmonisation and grading

Categorical fields use the controlled vocabularies in `schema.md`. Grading is mechanical, from the control-referenced ratio, by **CTCAE v5.0** cut-offs — a published standard, not thresholds invented here — and only for the readouts CTCAE defines. 918 of 2,685 rows are graded; the remaining 1,767 each state in `grade_basis` why no published rule applies. The one deviation, applying CTCAE's limit-of-normal ratios to a matched-control ratio, is named in every graded row's `grade_basis` so it cannot be overlooked. All grades are `provisional`.

8. Predictor and indicator variables

The variables themselves are defined in `schema.md`, their definitions are in the `data_dictionary` sheet of the workbook, and their ****distributions are reported in the narrative document**** (section 4), which is where the Challenge asks for them. They are computed from the tables at build time in both documents, never transcribed.

On negative controls. 640 rows carry a measured null (`effect_direction = no_change`) — the class most at risk of meaning "nobody looked". The extraction contract required a null only where the source reports a measured null, with reporting silence recorded separately in `notes`, and verification tested this stratum specifically because a prior review of the sibling kidney dataset found its negative class was substantially "nobody looked". **That defect does not repeat here:** four independent reviewers tried to break the null class and could not.

`control_class` records the reference each row was compared against. Vehicle, buffer and placebo rows dominate; the sequence-matched negative control — a scrambled or reverse-complement oligo, the only control that separates a sequence effect from a chemistry or formulation effect — is present on 8 rows. That number is small and is stated rather than smoothed over.

10. Quality control

Two committed scripts, both exiting non-zero on failure:

- `validate_dataset.py` — 112/112 structural checks (keys, referential integrity, vocabularies, grade reproducibility, sequence/modification consistency, roll-ups, and nine invariants added after verification). All pass. Defects caught during the build, and their fixes, are logged in `schema.md`.
- `verify_against_sources.py` — re-reads the committed documents and confirms every numeric readout appears in the source its row cites. **1,876 / 1,876 located.**

Structural QC proves internal consistency; source verification proves the numbers were not invented. Neither proves a number was read from the *right* cell, so a third pass did that by hand: 174 rows re-checked by reviewers instructed to refute them (117 confirmed, 50 corrected, 2 refuted, 5 unverifiable, no fabrication). Ten defect classes were found and corrected in the build; the residue is carried as open issues. Both the corrections and the residue are documented in `coagulopathy.md`.

11. Limitations

Stated in full in `README.md`. In brief: grades are provisional and mechanical; no clinical compound has a published sequence; PMO chemistry rests on regulatory silence rather than a measured null; prothrombotic rows come overwhelmingly from one compound family; the quantified PS class effect rests overwhelmingly on one study; figure-only values were not digitised.

12. Reproducibility

Deterministic and fully committed. From a clean checkout, with no network access:

`build_dataset.py` → `validate_dataset.py` → `verify_against_sources.py`. The build reads the committed extraction records in `sources/extraction/` and writes `data/`; it uses only the Python standard library. Every number in this document and in `README.md` is recomputable from `data/`.